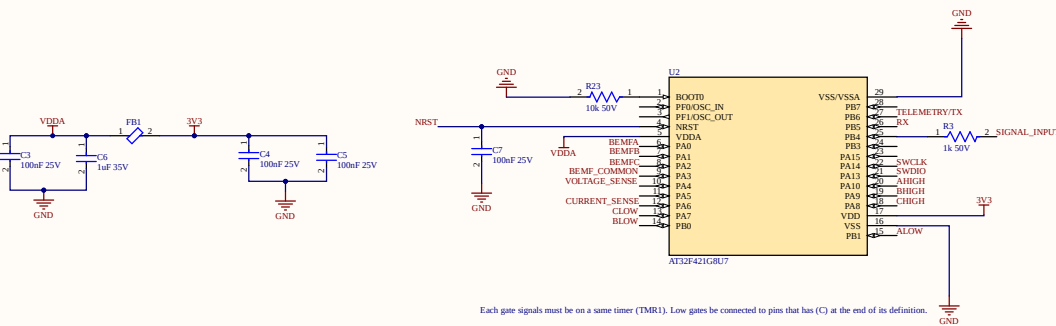
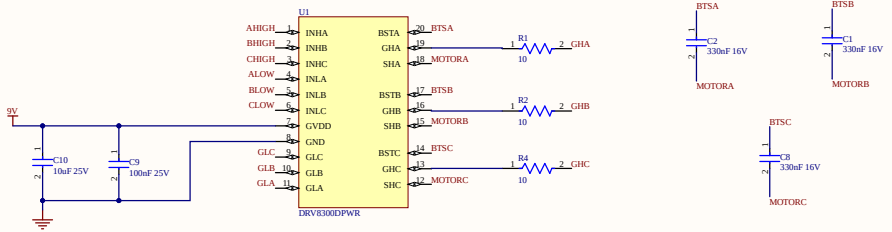


MCU (AT32F421G8U7)

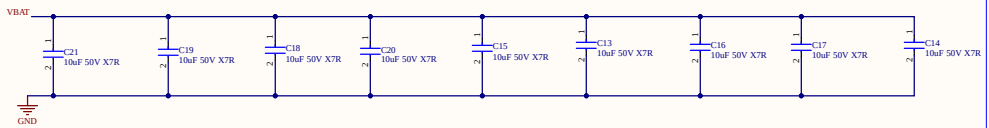


Each gate signals must be on a same timer (TMR1). Low gates be connected to pins that has (C) at the end of its definition.

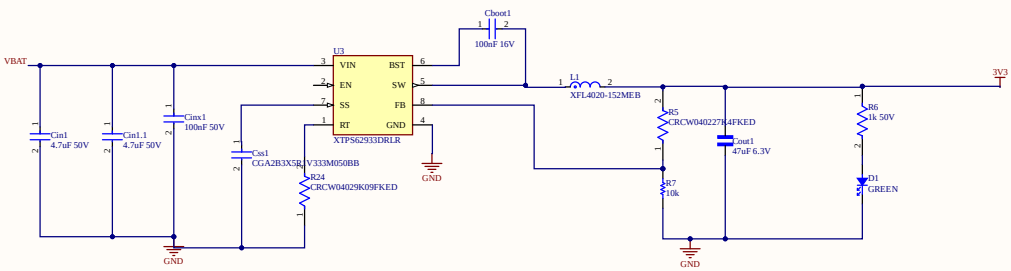
Power Driver Module



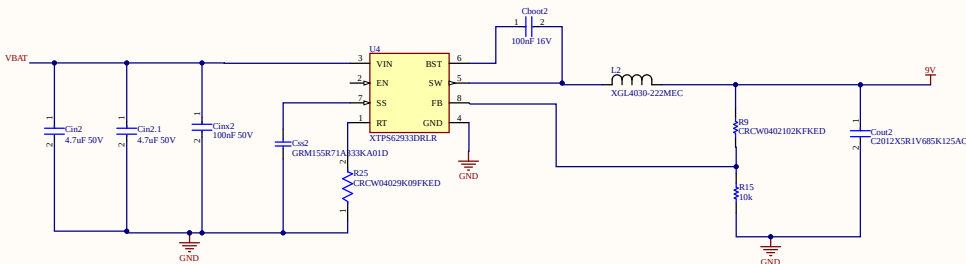
Decoupling Capacitors



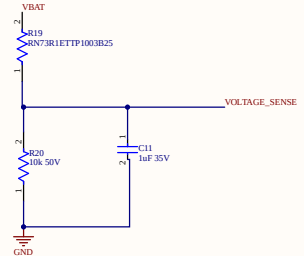
3V3 Buck



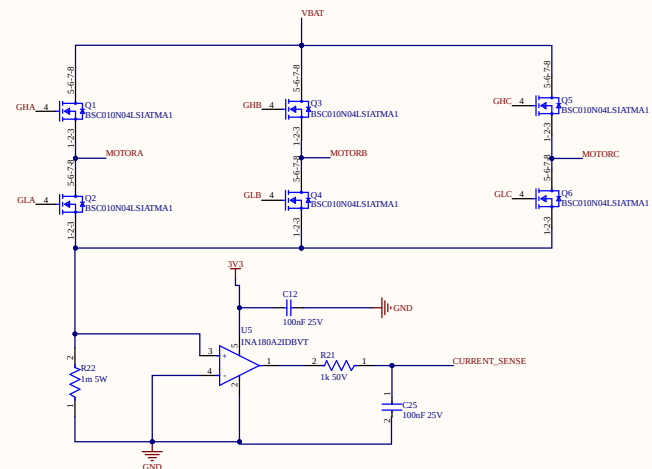
9V Buck



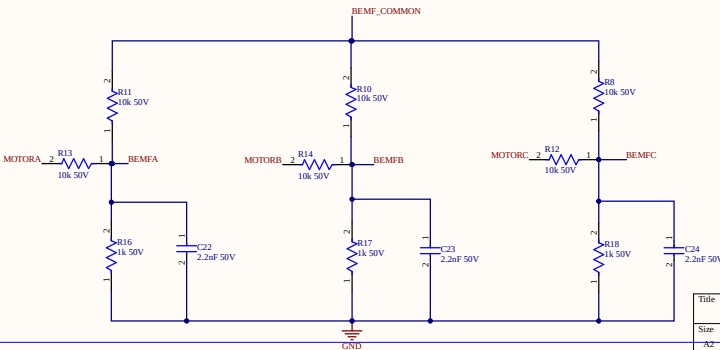
Voltage Sense

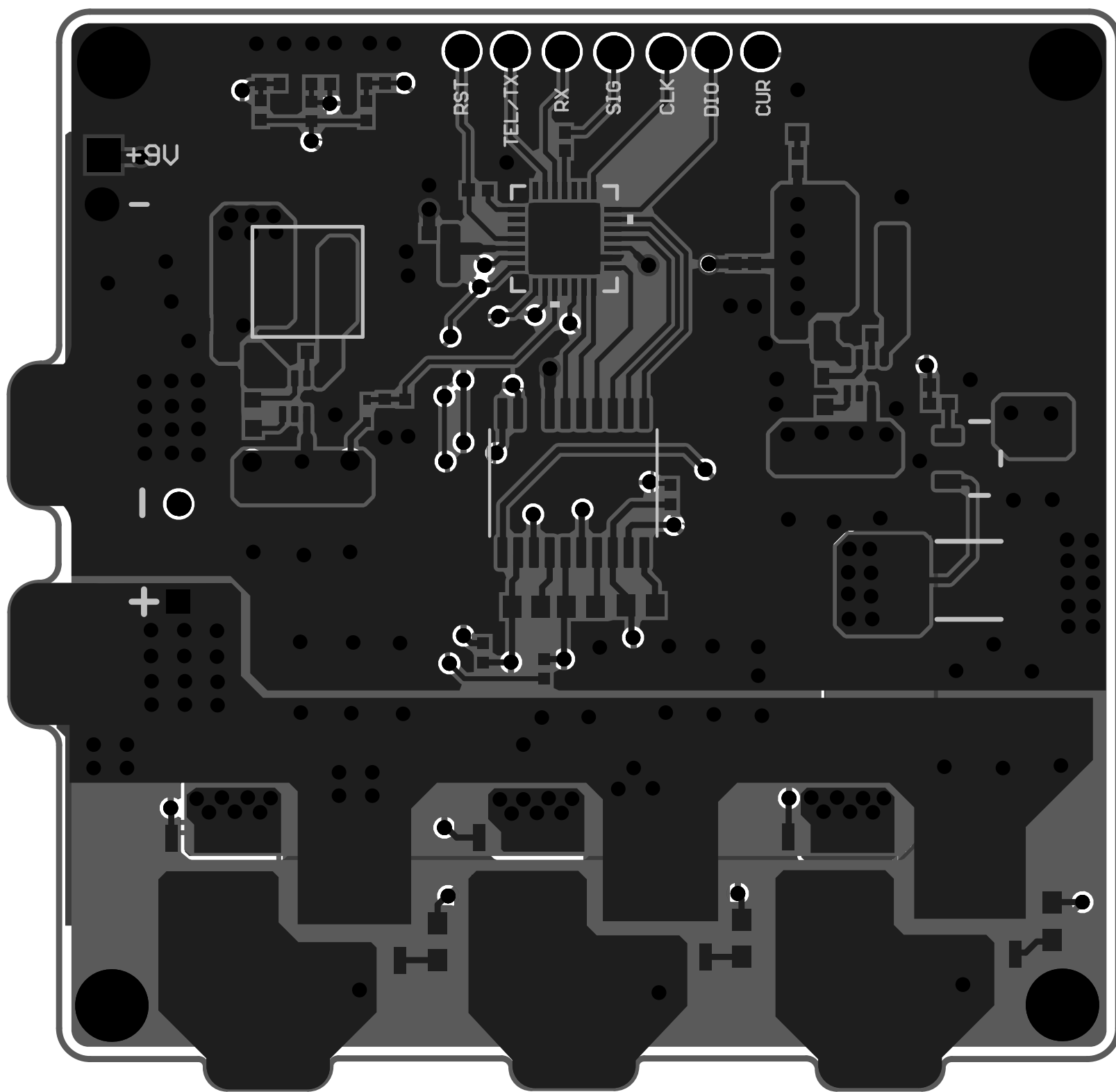


Mosfet Network



Back EMF Network





Comment	Description	Designator	Footprint	LibRef	Quantity
330nF16V	CAP CER0.33UF 16V X7R0603	C1, C2, C8	FP-C0603C-CF-MFG	OMP-03020-005049-1	3
100nF25V	CAP CER0.1UF 25V X5R0402	C3, C4, C5, C7, C9, C25	FP-C0402-BB-MFG	OMP-03020-014070-2	6
1uF35V	Ceramic Capacitor General Use 1uF±10% 35V X5S0402	C6, C11	FP-0402-L_1_0_0_2-W_0_5_0-MFG	OMP-14477-003658-1	2
10uF25V	CAP CER10UF 25V X5R0603	C10	FP-ZF818AF61E106MED1 L-MFG	OMP-06035-062385-2	1
100nF25V		C12	CAPC1608X87X35ML1 5T15	OMP-2006-03295-1	1
10uF50V X7R	CAP CER10UF 50V X7R1206	C13, C14, C15, C16, C17, C18, C19, C20, C21	FP-CL31B106KBHNNNE-MFG	OMP-2000-07530-2	9
2.2nF50V	Cap Ceramic 2.2nF 50V X7R±10% SMD 0402 +125°C Paper T/R	C22, C23, C24	FP-CL286-IPC_C	OMP-13271-000679-1	3
100nF16V	CAP CER0.1UF 16V X7R0402	Cboot1, Cboot2	FP-C0402-BB-MFG	OMP-2008-03082-2	2
4.7uF50V	CAP CER4.7UF 50V X5R0805	Cin1, Cin1.1, Cin2, Cin2.1	FP-CC0805-0_85-MFG	OMP-03422-000966-1	4
100nF50V	Cap Ceramic 100nF 50V X7R10% Pad SMD 0603 +125°C Automotive T/R	Cinv1, Cinv2	FP-CGA3E-0_3-0_1-IPC_A	OMP-2000-07546-2	2
47uF6.3V	47uF6.3V20% Multilayer Ceramic Capacitor X5R0603 (1608 Metric)	Cout1	CAPC1608X100N	MSAS168BB5476MPC A01	1
C2012X5R1V689K125 AC	Multilayer Ceramic Capacitors 6.8uF ±10% 35V X5R SMD 0805	Cout2	FP-C2012-125-0_2-IPC_B	OMP-08246-006695-1	1
CGA2B3X5R1V333M0 50BB	Cap Ceramic 33nF 35V X5R20% Pad SMD 0402 +65°C Automotive T/R	Css1	FP-CGA2B-IPC_A	OMP-08246-000404-1	1
GRM155R71A333KA01 D		Css2	CAPC1005X55X25N10 5T10	OMP-2008-04594-1	1
GREEN	LED YLW-GRN 0402 SMD	D1	LED0402G	VLJMG1500-GS08	1
BLM15BD601S2LD	Chip Ferrite Bead, 0402, 600Ω @ 100MHz, 0.65Ω, 25%, 200mA	FB1	FP-BLM15-0_05-IPC_B	OMP-06046-002695-1	1
XFL4020-152MEB	Power inductor, shielded, 1.5uH 20% 15.8mOhm 6.7A, SMT, RoHS, halogen free	L1	FP-XFL4020-MFG	OMP-11268-000205-1	1
XGL4030-222MEC	Description	L2	IND_XGL4030_COC	XGL4030-222MEC	1
BSC010N04LSATMA1	MOSFET N-CH 40V 37A/100A TDSO N	Q1, Q2, Q3, Q4, Q5, Q6	FP-PG-TDSO N-8-MFG	OMP-32901-000662-1	6
10	RES 10Ω 5.00% 0.1W 0603	R1, R2, R4	FP-AC0603-MFG	OMP-03412-039150-1	3
1k 50V	RES 1kΩ ±1% 0.063W 0402	R3, R6, R16, R17, R18, R21	FP-AC0402-IPC_C	OMP-03412-047873-1	6
CRCW040227K4FKED	RES Thick Film, 27.4kΩ, 1% 0.063W, 100ppm/°C, 0402	R5	FP-CRCW0402-e3-MFG	OMP-2002-08433-2	1
10k		R7, R15	RES 1005X40X25ML0 5T10	OMP-2002-03043-1	2
10k 50V	RES SMD 1.2K OHM 1%/10W 0402	R8, R10, R11, R12, R13, R14, R20, R23	FP-MCS0402-IPC_C	OMP-2002-02447-2	8
CRCW0402102KFKEED		R9	RES 1005X40X25ML0 5T05	OMP-1011-00737-2	1
RN73R1ETTP1003B25	100kΩ±0.1% 0.063W, 1/16W Chip Resistor 0402 (1005 Metric) Moisture Resistant Thin Film	R19	FP-RN73R1E-MFG	OMP-04015-069824-1	1
1m 5W	RES 0.001 OHM 1% 5W 2512	R22	FP-WSLF2512-0_43-MFG	OMP-02407-013110-1	1
CRCW04029K09FKED	RES Thick Film, 9.09kΩ, 1%, 0.063W, 100ppm/°C, 0402	R24, R25	FP-CRCW0402-e3-IPC_A	OMP-2002-00285-2	2
DRV8300DPMWR		U1	SOP65P640X120-20N	DRV8300DPMWR	1
AT32F421G8U7	Description	U2	QFN28 4x4 ART	AT32F421G8U7	1
XTPS62933DRLR	3.8-V to 30-V, 3-A, 200-kHz to 2.2-MHz, low-IQ synchronous buck converter in SOT-583 package 8-SOT-5X3-40 to 150	U3, U4	FP-DRL0008A-MFG	OMP-04918-000610-1	2
INA180A2IDBVT	IC CURR SENSE1 CIRCUIT SOT23-5	U5	FP-DBV0005A-MFG	OMP-04895-000174-1	1